

JEE MAIN REVISION MASTER: ATOMS AND NUCLEI 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 47 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Atoms And Nuclei - Core Application	47 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Atoms And Nuclei - Core Application (Very High Yield — 47 PYQs)

Core Principle: Key problem pattern in Kcet Atoms And Nuclei.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "An electron transition takes place from excited state to ground state in hydrogen atom, then..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Bohr's second postulate implies quantisation of..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The radius of first orbit in hydrogen atom is 5.3×10^{-11} m. The kinetic energy E_K , potential energy E_P and total energy..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

An electron transition takes place from excited state to ground state in hydrogen atom, then

- (A) Its kinetic energy increases but potential energy and total energy decrease
- (B) Its kinetic energy, potential energy and total energy decrease
- (C) Kinetic energy decreases, potential energy increases but total energy remains same
- (D) Kinetic energy and total energy decrease but potential energy increases

Correct Answer: None

Solution Analysis: For hydrogen atom in the Bohr model, $E_n = -\frac{13.6}{n^2} \text{ eV}$ where E_n is the total energy of the electron in the n^{th} orbit. When the electron comes from an excited state to the ground state, n decreases. For example, from $n=2$ to $n=1$: $E_2 = -3.4 \text{ eV}$ $E_1 = -13.6 \text{ eV}$ So the total energy becomes more negative. Hence, total energy decreases. For an electron in a hydrogen atom, $K = -E$ and $U = 2E$ where K is kinetic energy and U is potential energy. Since total energy E decreases and becomes more negative: Kinetic energy $K = -E$ increases. Potent